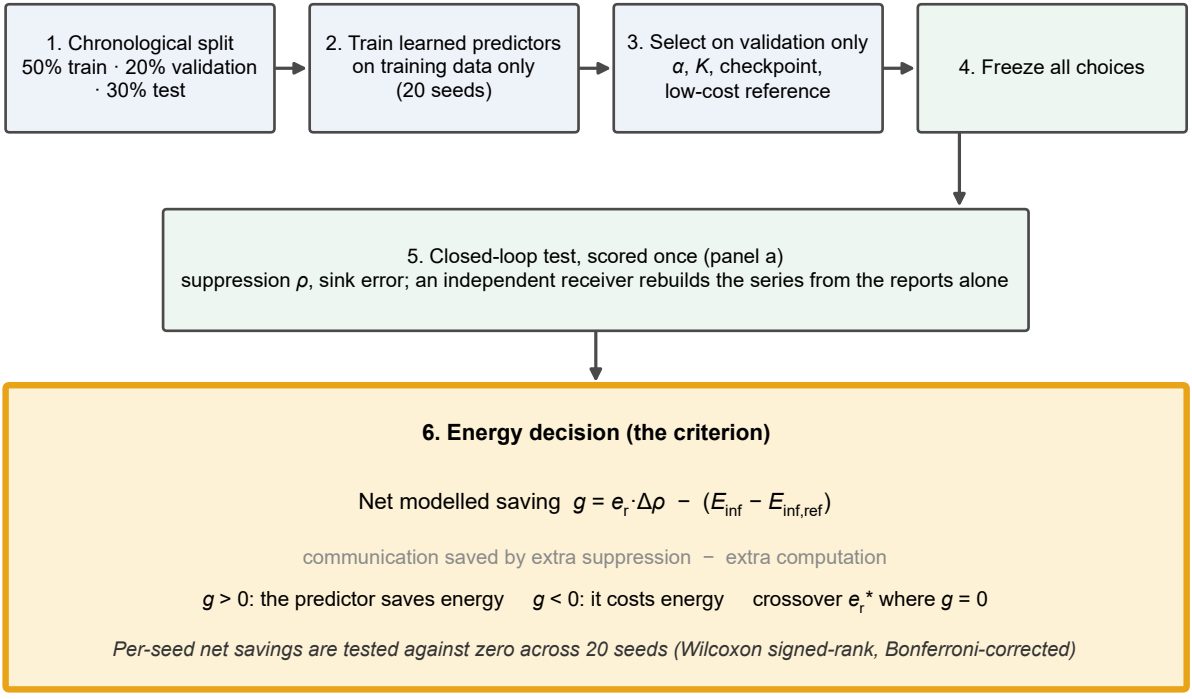
(b)


1. Chronological split
50% train · 20% validation
· 30% test
2. Train learned predictors
on training data only
(20 seeds)
3. Select on validation only
α, K, checkpoint,
low-cost reference
4. Freeze all choices
5. Closed-loop test, scored once (panel a)
suppression ρ, sink error; an independent receiver rebuilds the series from the reports alone
6. Energy decision (the criterion)
Net modelled saving g = e_r·Δρ − (E_inf − E_inf,ref)
communication saved by extra suppression − extra computation
g > 0: the predictor saves energy
g < 0: it costs energy
crossover e_r* where g = 0
Per-seed net savings are tested against zero across 20 seeds (Wilcoxon signed-rank, Bonferroni-corrected)